

PATHI MANIKANTA

Research Engineer | Code Generation & Model Evaluation | AI Trainer & Reviewer

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PROFESSIONAL SUMMARY

Research-oriented engineer who evaluates and improves AI systems for a living. I review how models generate code, catch where they hallucinate or break, and build the rubrics and verifiers that keep output quality honest. I write Python daily and work across Java, C, Go, and TypeScript when the task calls for it. Comfortable debugging unfamiliar codebases, designing evaluation criteria from scratch, and shipping clean, well-tested code under tight deadlines.

SKILLS

AI Evaluation: LLM evaluation, AI red teaming, RLHF, prompt engineering, rubric design, agent trajectory review, multimodal assessment, fact-checking, data annotation, hallucination detection, AI safety testing

Programming: Python (strong), Java, C, Go, JavaScript, TypeScript, SQL

Problem Solving: Data structures, algorithms, competitive programming, debugging, analytical thinking

Machine Learning: scikit-learn, TensorFlow, Keras, NumPy, pandas, NLTK, NLP, cross-validation, data-leakage detection

Libraries and Frameworks: React, pytest, Streamlit, SHAP

Tools and Testing: deterministic verifiers, Git, GitHub, GitHub Actions, Jupyter Notebook

Databases: SQL, MongoDB

WORK EXPERIENCE

AI Trainer and Reviewer (Freelance)	Jun 2026 – Present
Outlier (Scale AI)	India

- Earned promotion from Attempter to Reviewer within 3 months for evaluation accuracy, guideline compliance, and consistent task quality.
- Authored 50+ pass/fail rubrics with 15 to 25 criteria each and built pytest verifiers to validate code generation and task outputs.
- Reviewed 100+ agentic trajectories for tool selection, sequencing, argument accuracy, hallucinations, and instruction-following failures across diverse codebases.
- Developed challenging prompts, reference trajectories, and corrective nudges for OpenClaw Atlas and Blue Shell multimodal evaluations, supported by rubrics and written justifications.
- Assessed model safety for Lobster Safety, classified failure types and response severity, and documented technical decisions against project guidelines.
- Collaborated with cross-functional review teams to set evaluation standards and ensure consistent, high-quality model assessment outputs.

Software and Machine Learning Engineer (Freelance)	Sep 2026 – Present
Tensium	Remote

- Author CPU-based, long-horizon machine learning tasks requiring multistep reasoning, code generation, and execution across Python workflows.
- Develop, debug, and refactor Python code for software and machine learning pipelines, improving maintainability and performance.

AI Trainer, LLM Evaluation (Freelance)	Jul 2026 – Present
Handshake AI	India

- Authored challenging, single-answer multihop research questions with reference answers to evaluate frontier model performance and accuracy.
- Built automated submission quality checks to validate task packages against rejection rules, ensuring output consistency before submission.

MCX Advisor	2025 – 2026
Zebu Share and Wealth Management	Chennai, India

- Supported client trading operations through order processing, position monitoring, and account and profit-and-loss updates.

PROJECTS

JudgeLab – Judge Reliability Lab | GitHub

Python, Streamlit, statistics

- Benchmarked GPT-4 judgments against 3,355 human expert votes from MT-Bench, measuring Cohen's kappa of 0.46 on decisive verdicts.
- Measured 15.8% order-swap inconsistency and analyzed verbosity bias and per-category reliability to identify limitations in automated judging.

TrajLens – Agentic Trajectory Auditor | GitHub

Python, Streamlit

- Built an auditor that grades 14 recorded tool-use traces against a 9-criterion rubric, including checks for buried errors and unauthorized actions.
- Compared verifier results with human gold labels using Cohen's kappa; identified hallucinated-value and retry-loop failures missed by a results-only LLM judge.

EvalGate – CI/CD Quality Gate | GitHub

Python, GitHub Actions, scikit-learn

- Built a CI/CD pipeline that runs 11 tests, retrains the model, and enforces ROC-AUC and cross-validation drift checks before promoting artifacts. Refactored pipeline code for maintainability.
- Automated deployment of a model-quality dashboard to GitHub Pages with weekly scheduled retraining.

Customer Churn Predictor | GitHub

scikit-learn, SHAP, Streamlit

- Deployed a Gradient Boosting pipeline on 7,043 IBM Telco records, achieving 0.843 test ROC-AUC and 0.847 +/- 0.013 across five-fold cross-validation.
- Added SHAP explanations and a Streamlit interface to connect churn predictions with customer-retention actions.

Diabetes Foot Pressure Classification | GitHub

Arduino, scikit-learn, Keras

- Corrected session leakage using GroupKFold, revising reported accuracy from 99% to 92.5% for a more reliable evaluation.

Spam Email Classifier | GitHub

Python, NLTK, TF-IDF, SVM

- Built an NLP pipeline with NLTK and TF-IDF, achieving 97.8% accuracy with a Linear SVM on 1,000+ emails.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering (Data Science)

2021 – 2025

Jansons Institute of Technology (Autonomous), affiliated with Anna University

Coimbatore, India

CGPA: 8.0/10

CERTIFICATIONS

- Problem Solving (Intermediate) – HackerRank
- Frontend Developer (React) – HackerRank
- Introduction to Agents – Mercor (2026)
- Prompt Engineering (OutlierEDU) – Outlier AI (2026)
- Claude Code 101 – Anthropic (2026)
- AI Fluency for Builders – Anthropic (2026)
- AWS Academy Cloud Foundations – Amazon Web Services (2022)
- Ethical Hacking and Penetration Testing – Scholiverse
- Microsoft 365 Productivity Advanced – Microsoft (2022)
- Enrolled Agent (EA) credential – in progress